

Five Business Days: Disclosure Deadlines, Activist Stakes, and Takeover Gains

Research Proposal

Junyu (Austin) Li*

August 2026

Abstract

In February 2024, the Securities and Exchange Commission shortened the initial Schedule 13D filing deadline from ten calendar days to five business days and declined to extend beneficial-ownership reporting to certain cash-settled derivatives. This paper examines whether the reform changed activist campaign formation, the composition of activist positions, and the timing of takeover gains. The empirical design uses a predetermined target-level exposure measure: the number of trading days required to acquire an additional five percentage points of ownership at a participation cap. This measure captures how tightly the shorter deadline constrains undisclosed post-threshold accumulation. I use it to test whether campaign entry, target composition, stakes at filing, and Item 6 derivative disclosure respond more strongly for more exposed targets. I also decompose target gains into the filing return, the post-filing run-up, and the bid markup, holding the trigger-to-bid horizon fixed. A parsimonious model of accumulation over a filing window derives the exposure measure and identifies the conditions under which a shorter deadline reduces the stake acquired before disclosure. The design reports pre-specified minimum detectable effects and uses the SEC's published tables as a pre-reform benchmark. It uses realised behaviour to evaluate the SEC's projected \$810 million annual loss in shareholder value.

1 Institutional Setting and Research Question

In February 2024, the Securities and Exchange Commission shortened the initial Schedule 13D filing deadline from ten calendar days to five business days, the first change to the accumulation window since 1968 (U.S. Securities and Exchange Commission 2023). Once an investor crosses five per cent beneficial ownership, the amendment reduces the period in which further shares may be acquired before the position is disclosed. The final rule also declined to adopt proposed Rule 13d-3(e), which would have treated certain cash-settled derivatives as beneficial ownership. The reform therefore changed both the period for undisclosed equity accumulation and the relative attractiveness of derivative exposure.

The paper asks whether this change altered activist campaign formation, stake formation, and the distribution of takeover gains. It addresses three questions. First, did campaign entry or target composition change more for targets on which the deadline was more restrictive? Second, conditional on a campaign, did reported stakes and the use of cash-settled derivatives

*UCL.

change with that exposure? Third, did activist-related takeover gains appear at filing, during the pre-bid run-up, or in the announced bid markup?

Exposure is measured by the required trading days,

$$\text{RTD}_i = \frac{0.05 \times S_{\text{out},i}}{\phi \times \text{ADV}_i}, \quad (1)$$

the number of trading days required to acquire an additional five percentage points of ownership in target i at participation cap ϕ . Here $S_{\text{out},i}$ denotes shares outstanding and ADV_i is average daily volume, measured in a lagged pre-trigger window. Since the statutory clock begins only at the five-per-cent crossing, the deadline limits the undisclosed increment beyond that threshold to $0.05 \times d/\text{RTD}_i$ percentage points. Higher RTD_i therefore indicates a tighter constraint for a given legal window d , without using the campaign’s realised accumulation or outcome.

2 Economic Mechanism and Predictions

A shorter filing window affects activist activity through two margins. It raises the price-impact cost of acquiring a given undisclosed increment, and it caps the increment that can be acquired before disclosure. These effects may discourage campaigns or shift them towards more liquid targets, larger activists, or different campaign objectives. They may also encourage substitution from common-stock accumulation to cash-settled derivatives reported in Item 6, an instrument that proposed Rule 13d-3(e) would have covered but the final rule did not.

The economic magnitude is uncertain. The SEC projected approximately \$810 million a year in foregone shareholder value, reasoning that less time for undisclosed accumulation would deter value-increasing campaigns (U.S. Securities and Exchange Commission 2023); Polk et al. (2024) project lower activist returns using historical data. At the same time, the SEC’s tables indicate that roughly 80 per cent of filers had already completed accumulation by the five-business-day deadline and that the rule binds materially for only a minority of campaigns. The central empirical question is consequently not whether the reform must chill activism, but whether any response is concentrated among targets for which the deadline plausibly binds.

The interpretation of takeover premia is likewise not mechanical. Celentano and Levine (2025) find lower conditional bid premia for activist targets and relate this result to bargaining. Yet the pre-bid price may already reflect information revealed at filing and during subsequent trading. Schwert (1996) finds little substitution between run-ups and markups, whereas Betton et al. (2014) finds stronger substitution after adjusting the run-up for changes in stand-alone value. I therefore decompose total target gain into the filing return $G_{\text{fil}} = \log(P^F/P^0)$, the post-filing run-up $G_{\text{run}} = \log(P^B/P^F)$, and the bid markup $G_{\text{mkp}} = \log(B/P^B)$. The decomposition is evaluated over a fixed trigger-to-bid horizon so that variation in event timing cannot mechanically generate the predicted pattern.

3 Contribution

The contribution is a realised evaluation of the February 2024 Schedule 13D acceleration using predetermined variation in required trading days. Existing disclosure-threshold models study stake-size triggers; models of activist intermediation and bidder solicitation take the toehold as given or make disclosure instantaneous; and structural work does not vary the filing window. The model here makes the legal deadline a time constraint on post-threshold accumulation and yields

the exposure measure used in the empirical design. The paper also tests whether the reform affected derivative use and records the timing of activist-related takeover gains. It does not claim to re-establish the activist-as-intermediary mechanism or the capitalisation interpretation of conditional premia. Those results instead motivate the empirical tests.

4 A Deadline Model of Activist Stake Formation

The model provides the economic framework for the empirical evaluation. Its main purpose is to derive the target-level exposure measure RTD_i and to show when the statutory filing deadline limits post-trigger accumulation. It also clarifies two downstream empirical questions: whether earlier disclosure attracts or deters bidders, and whether changes in measured takeover premia represent a reallocation of gains across event time or a change in total target-shareholder value.

The distinctive margin is the time available to accumulate shares after crossing the five-per-cent threshold. Existing disclosure models study a stake-size threshold (Ordonez-Calafi and Bernhardt 2022); they do not treat the filing window as a deadline in an accumulation problem. Likewise, Celentano and Levine (2025) provide the capitalisation interpretation of a lower measured takeover premium, while Betton et al. (2014) provide the relevant evidence on substitution between run-ups and markups. The model instead connects the legal time limit to stake formation, bidder entry, and the empirical decomposition of target gains.

4.1 Setting and timing

A risk-neutral activist considers a campaign against target i , which has $S_{\text{out},i}$ shares outstanding and liquidity L_i . In the empirical implementation, liquidity is measured by average daily volume over the six months ending 60 trading days before the trigger, with Amihud illiquidity as an alternative. The lag keeps the activist's untimed toehold purchases out of the measurement window and aligns the liquidity measure with the unaffected-price anchor. All agents discount at rate $\delta \in (0, 1)$ per period.

At t_0 , the activist privately observes an opportunity $\theta \geq 0$, the per-share value improvement available if the campaign succeeds. The activist then builds a toehold and crosses the five-per-cent beneficial-ownership threshold at t_1 . This pre-trigger phase is not subject to the filing deadline because the Section 13(d) clock begins only after the threshold is crossed (U.S. Securities and Exchange Commission 2023).

The crossing date is the trigger date observed in the data. The unaffected price P^0 is measured a fixed k trading days before that date. This anchor is necessary because accumulation itself can move the price. Slow toehold building before trigger-minus- k may still contaminate the measure, so the empirical analysis varies k across 30, 60, and 120 trading days.

Between the trigger and filing, the activist can add an undisclosed increment Δx over at most d business days. The filing stake is $x = 0.05 + \Delta x$. Before the reform, ten calendar days corresponded to approximately seven business days; after the reform, $d = 5$. The reform therefore affects only the post-trigger increment.

Filing occurs at the earlier of the statutory deadline and voluntary disclosure. Early filing remains feasible and may be optimal, so the deadline binds only for an endogenous subset of campaigns. Let P^F denote the post-filing price and define $G_{\text{fil}} \equiv \log(P^F/P^0)$. Potential acquirers subsequently search and evaluate the target. Let P^B denote the bid-eve price and $G_{\text{run}} \equiv \log(P^B/P^F)$. If bargaining produces a transaction, the final offer is B and $G_{\text{mkp}} \equiv \log(B/P^B)$; otherwise, the firm remains freestanding.

The activist's private information is summarised by the scalar θ . The model does not introduce a separate operational-versus-sale type. With a two-dimensional type and a two-dimensional signal (x , Item 4 purpose), the filing generically separates types and P^F collapses to the full-information price, eliminating the signal noise needed elsewhere in the model. Operational and sale intent therefore enter only as empirical Item 4 heterogeneity in the entry elasticities below, not as a second signalling dimension. Section 4.8 discusses this restriction.

4.2 The cost of accumulating before the deadline

The activist trades against competitive dealers with linear price impact $\lambda_L(L)$, where λ_L falls with target liquidity. Empirically, λ_L is proportional to inverse average daily volume or to Amihud illiquidity. Detection risk and dealer inventory capacity limit purchases to a fraction ϕ of average daily volume per day.

At daily purchase rate r , excess execution cost is $(\lambda_L/2)r^2$. Convexity makes a constant rate optimal over a fixed window. The cost of acquiring x over d days is therefore

$$C(x; d, L) = \frac{\lambda_L(L)}{2} \frac{x^2}{d}. \quad (2)$$

Its relevant derivatives are

$$C_x = \frac{\lambda_L x}{d} > 0, \quad C_{xx} = \frac{\lambda_L}{d} > 0, \quad C_{xd} = -\frac{\lambda_L x}{d^2} < 0, \quad (3)$$

and the horizon scaling is exact: $C(x; \kappa d, L) = \kappa^{-1} C(x; d, L)$ for every $\kappa > 0$.

A longer window lowers both total and marginal accumulation cost. The negative cross-partial C_{xd} is derived from the trading technology rather than assumed. It is strongest for illiquid targets, for larger desired increments, and close to the deadline. Moving from approximately seven to five business days raises the quadratic cost of a fixed post-trigger increment by $7/5 = 1.4$, or 40 per cent. The same price-impact mechanism produces a pre-filing run-up proportional to $\lambda_L x/d$, linking the split between pre-filing leakage and the filing return to target liquidity.

Equation (2) applies only to the post-trigger increment Δx . The pre-trigger toehold is acquired over an unconstrained horizon chosen by the activist, so its cost does not depend on d and drops out of the comparative statics.

This cost is a static reduced form for a dynamic trading problem. A deadline version of the Kyle framework in Collin-Dufresne and Fos (2016) would make the intensity path endogenous and allow the activist to trade off price impact against information leakage. The strategic benchmark is Back et al. (2018). Those models add the shape of within-window trading, but not the sign of C_{xd} or the $1/d$ scaling whenever per-period impact is convex. Collin-Dufresne and Fos (2015) establish that informed trading during the Schedule 13D window is empirically important, which supports treating the window as a trading horizon; Kyle (1985) supplies the linear-impact primitive.

4.3 Exposure to the deadline and the stake at filing

The daily participation limit implies that target i can accommodate at most

$$\overline{\Delta x}_i(d) = \frac{\phi \text{ADV}_i d}{S_{\text{out},i}} \quad (4)$$

of undisclosed post-trigger accumulation. Define required trading days as

$$\text{RTD}_i = \frac{0.05 \times S_{\text{out},i}}{\phi \times \text{ADV}_i}, \quad \text{so that} \quad \overline{\Delta x}_i(d) = 0.05 \frac{d}{\text{RTD}_i}. \quad (5)$$

This is the number of trading days required to add five percentage points of ownership at the participation cap. Let $x_i^*(d) = 0.05 + \Delta x_i^*(d)$ denote the stake at filing.

Proposition 1 (Deadline exposure and stake formation). *Let $\Delta x^u(d)$ solve the unconstrained post-trigger problem*

$$\max_{\Delta x \geq 0} \{b(0.05 + \Delta x, d) - C(\Delta x; d, L) - F\},$$

where b is the gross benefit from the filing stake and F is a fixed campaign cost. The realised increment is

$$\Delta x_i^*(d) = \min\{\Delta x^u(d), \overline{\Delta x}_i(d)\}.$$

First, the deadline does not constrain the activist's ability to reach the disclosure threshold because the statutory clock starts at the crossing. It constrains only ownership acquired beyond the threshold. The maximum filing stake is therefore

$$0.05 \left(1 + \frac{d}{\text{RTD}_i}\right).$$

Second, for a desired increment Δx^u , the constrained set is

$$\left\{i : \text{RTD}_i > \frac{0.05d}{\Delta x^u}\right\}.$$

This set expands, in the set-inclusion sense, as d falls. On the constrained set, the shadow value

$$b_x(0.05 + \overline{\Delta x}_i(d), d) - C_x(\overline{\Delta x}_i(d); d, L) \geq 0$$

increases with RTD_i . Under the doubling calibration $\Delta x^u = 0.05$, a scale motivated by the pre-disclosure accumulation documented by Collin-Dufresne and Fos (2015), the constrained set is exactly $\{i : \text{RTD}_i > d\}$.

Third, suppose $b(\cdot, d)$ is C^2 and the objective is strictly concave in Δx , so $b_{xx} - C_{xx} < 0$, with derivatives of b evaluated at $x = 0.05 + \Delta x^*$. In the unconstrained region,

$$\frac{d \Delta x^*}{dd} = -\frac{b_{xd} - C_{xd}}{b_{xx} - C_{xx}} > 0 \quad \Longleftrightarrow \quad b_{xd} > C_{xd} = -\frac{\lambda_L \Delta x^*}{d^2}. \quad (6)$$

The filing stake moves one-for-one: $dx^*/dd = d \Delta x^*/dd$. In the constrained region, $\Delta x_i^* = \overline{\Delta x}_i(d)$ and

$$\frac{d \Delta x_i^*}{dd} = \frac{\phi \text{ADV}_i}{S_{\text{out},i}} > 0$$

without any additional restriction.

Consequently, the reform dose is a monotone function of RTD_i/d . The continuous dose is the primary model object. The indicator $\mathbf{1}\{\text{RTD}_i > d\}$ is the corresponding binary classification under the doubling calibration, not a general treatment definition.

Condition (6) is the model’s single-crossing restriction. A longer window delays filing and hence certification, which may reduce the probability that a bid arrives while the activist remains positioned. Thus b_{xd} may be negative. When it is, the condition requires $|b_{xd}| < \lambda_L \Delta x^* / d^2$. The restriction is weakest where the accumulation friction is largest: high λ_L , low d , and high Δx . It is unnecessary at the constrained corner. The sign is therefore conditional in the interior and unconditional only on the constrained set.

The proposition gives an economic interpretation to the accumulation ladder in the adopting release (U.S. Securities and Exchange Commission 2023). Roughly 80 per cent of filers complete their reported stake within five business days and belong to the slack set for their realised desired increment. The 3 per cent of non-corporate-action campaigns that remain below 90 per cent completion at the amended deadline form the deeply constrained tail. Equations (4) and (5) derive the exposure measure from lagged target-level, price-free inputs. The measure is defined for one-time filers and permits activist-by-quarter fixed effects.

4.4 Bidder entry

Potential acquirers are ex ante identical. Each pays search and evaluation cost s to learn its valuation, and entry continues until expected profit is zero. Earlier disclosure has two opposing effects.

A Schedule 13D filing certifies that a large, sophisticated holder has identified the target and is unlikely to defend incumbent management. The filing therefore performs part of the bidder’s screening and lowers s . This is the solicitation and search mechanism of Corum and Levit (2019) and the activist-intermediary mechanism of Burkart and Lee (2022), both taken as given here.

Disclosure can also deter entry by increasing the expected acquisition price. The disclosed stake and the associated engagement are capitalised into P^F , and a larger, better-informed bloc may bargain more aggressively. This is the pre-emption channel in Fishman (1988) and the disclosure-reform evidence in Bonetti et al. (2020). It also contains the negative entry effect in the companion model.

Let $h(s, B^e)$ be the free-entry bid hazard, decreasing in search cost and the expected bid price. Let e denote the earliness of disclosure, equal to minus the filing lag, holding the campaign fixed. Define

$$\varepsilon_{\text{cert}} = \left(-\frac{\partial \log h}{\partial \log s} \right) \left(-\frac{\partial \log s}{\partial e} \right), \quad \varepsilon_{\text{det}} = \left(-\frac{\partial \log h}{\partial \log B^e} \right) \left(\frac{\partial \log B^e}{\partial e} \right).$$

Both elasticities are non-negative, and

$$\frac{d \log h}{de} = \varepsilon_{\text{cert}} - \varepsilon_{\text{det}}. \tag{7}$$

Earlier disclosure raises the bid hazard if and only if $\varepsilon_{\text{cert}} > \varepsilon_{\text{det}}$. The model therefore does not impose a sign on bidder entry.

Certification is more likely to dominate when screening is costly and the counterparty set is unclear: opaque or hard-to-value assets, few natural strategic buyers, cold industry M&A markets, and Item 4 language indicating sale or strategic-alternatives intent. Deterrence is more likely to dominate when acquirers are already searching, as in hot merger markets or industries with an obvious set of strategic buyers.

The reform does not change disclosure timing alone. By reducing x^* through the preceding proposition, it can also lower B^e and thereby increase entry. This third effect partly offsets deterrence, so an empirical entry response cannot be attributed entirely to certification.

4.5 Bargaining and appropriability

Conditional on bidder entry, the final offer divides the surplus between the acquirer's stand-alone valuation V_a and the value Y under a successful campaign:

$$B = V_a + \beta(x)(Y - V_a), \quad \beta(x) = \bar{\beta}\lambda_{\text{app}}, \quad \lambda_{\text{app}} \in [0, 1]. \quad (8)$$

The restriction on β preserves the free-rider problem in Grossman and Hart (1980b). An unrestricted increasing $\beta(x)$ would assume that problem away. The analysis therefore imports only the conclusion of the companion model's tender-game appendix:

$$\lambda_{\text{app}} = 1 - q(1 - \gamma)\psi,$$

where q is the probability that a fringe raider appears at the disagreement node, $\gamma \in [0, 1]$ is the portability of the activist's improvement under a different owner, and $\psi \in [0, 1]$ measures pivotality. Appropriability falls with fringe-raid risk and rises with portability.

Each primitive has an observable counterpart. Fringe M&A intensity measured at the industry-year level proxies for q . The mix of strategic and financial acquirers, together with asset redeployability, proxies for γ . State and charter antitakeover provisions index charter dilution, while the bloc's stake relative to control thresholds indexes pivotality. Moreover, λ_{app} rises discontinuously when the bloc crosses the blocking threshold $1 - \tau_c$. This produces a stake-size threshold prediction linked directly to the deadline through x .

The tender game itself is not reproduced. Doing so would add an entry-cost distribution, a synergy distribution, charter dilution, and a control threshold as four additional primitives. To avoid a notation conflict, charter dilution, denoted ϕ in the companion appendix, is denoted ϕ_c here; ϕ continues to denote the daily participation cap.

4.6 Accounting for target gains

Total target gain between the unaffected price and the final offer decomposes exactly as

$$G_{\text{tot}} \equiv \log \frac{B}{P^0} = \underbrace{\log \frac{P^F}{P^0}}_{G_{\text{fil}}} + \underbrace{\log \frac{P^B}{P^F}}_{G_{\text{run}}} + \underbrace{\log \frac{B}{P^B}}_{G_{\text{mkp}}}. \quad (9)$$

This is an accounting identity. Its empirical content comes from the restrictions it places on changes in the three components.

Suppose the bid-eve price is the discounted expected resolution value,

$$P^B = \delta \left[\varrho B + (1 - \varrho)P^{\text{fail}} \right],$$

where $\varrho \in (0, 1]$ is the completion probability and $P^{\text{fail}} < B$ is the freestanding value. Then

$$G_{\text{mkp}} = \log(1/\delta) - \log \left(\varrho + (1 - \varrho) \frac{P^{\text{fail}}}{B} \right) \geq \log(1/\delta) > 0.$$

In simple returns, the measured markup is bounded below by $(1 - \delta)/\delta$ plus a strictly positive deal-risk term. A fully anticipated deal therefore still has a positive measured markup; a declining markup alone has no structural interpretation.

Holding the transaction fixed, so that B and P^0 do not change, equation (9) implies the testable restriction

$$dG_{\text{mkp}} = -(dG_{\text{fil}} + dG_{\text{run}}). \quad (10)$$

Under pure capitalisation, gains move across the three components while G_{tot} remains invariant. If the reform destroys target value, G_{tot} falls.

This restriction separates capitalisation from value destruction in the data. Celentano and Levine (2025) give the capitalisation interpretation in their structural counterfactual. Betton et al. (2014) document substitution between value-adjusted run-ups and markups, whereas Schwert (1996) find little substitution without that adjustment. The empirical analysis therefore tests the restriction on event-time-invariant components. A naive calendar-window comparison would mechanically reassign days between the filing and run-up components when the filing deadline changes, producing a truncation artefact that runs opposite to the prediction being tested.

4.7 Welfare and empirical sufficient statistics

Let $W(d)$ denote expected target-shareholder gain. In the interior, the envelope theorem removes the indirect effect of d through x^* from the activist's own payoff. Any surviving dx^*/dd term is therefore an externality on target shareholders. At the constrained corner, the activist's payoff changes at first order with the shadow value of the filing window, and the stake response is $dx^*/dd = \phi \text{ADV}_i / S_{\text{out}}$.

The welfare decomposition is

$$\begin{aligned} \frac{dW}{dd} = & \underbrace{\frac{\partial \mathbb{P}(\text{bid})}{\partial d} \mathbb{E}[G_{\text{tot}} \mid \text{marginal deal}]}_{\text{(i) entry margin}} + \underbrace{\frac{\partial \mathbb{P}(\text{camp})}{\partial d} \mathbb{E}[G_{\text{tot}} \mid \text{marginal campaign}]}_{\text{(ii) campaign-supply margin}} \\ & + \underbrace{\mathbb{P}(\text{bid}) \frac{\partial \mathbb{E}[G_{\text{mkp}}]}{\partial \beta} \beta'(x^*) \frac{dx^*}{dd}}_{\text{(iii) stake-bargaining margin}}. \end{aligned} \quad (11)$$

The entry margin has the sign of equation (7) and must be estimated. The campaign-supply margin is non-negative because a longer window supports more campaigns. Since the reform has $\Delta d < 0$, this term enters the welfare effect negatively and captures chilling. The stake-bargaining margin operates through the filing stake: a shorter window lowers x^* , which lowers $\beta(x^*)$ and the conditional markup independently of entry and campaign supply.

The conditional expectations in equation (11) concern marginal, not average, units. Marginal deals are absent under the counterfactual regime, and deterred campaigns are unobserved. Replacing these objects with the average observed premium would substitute an ATE for a LATE. The intensity design addresses this problem because campaigns with high RTD_i are the units at the margin of being constrained or deterred. Its exposure-based estimands therefore correspond to the weights required by the welfare decomposition.

Finally, accumulation can move the target's price before filing. Part of G_{fil} is consequently a transfer from selling shareholders to the activist rather than a gain to a fixed shareholder cohort. Welfare is therefore defined for shareholders present at trigger-minus- k , and the transfer component is reported separately.

4.8 Scope of the model

The filing price is a Bayesian update based on an exogenously garbled filing. Nature adds noise to the disclosed $(x, \text{purpose})$ pair, and a false Item 4 statement carries an exogenous securities-liability cost. The activist’s stake is not solved as an endogenous signal. A continuous signal chosen by a privately informed type can support pooling, partial-pooling, and separating equilibria. Without an equilibrium refinement and a uniqueness result, P^F would not be a well-defined empirical object.

This restriction also avoids a separate closure problem. If the final offer B in equation (8) is independent of P^F , the capitalisation result follows mechanically because the denominator moves while the numerator does not. If B instead depends on P^F through a market-price floor, the mapping $P^F \rightarrow B \rightarrow P^F$ becomes a fixed point and requires a contraction condition. The model therefore holds B independent of P^F and presents the invariance result as a measurement restriction rather than the paper’s main theoretical claim. Endogenising the filing price is left for separate work.

Other deliberate omissions are the operational-versus-sale distinction as a second private type, the tender game underlying λ_{app} , bidder competition and repeated fringe entry, and the within-window trading path. Cash-settled derivatives are treated in the empirical Item 6 analysis as an outside option but are not priced in the model. This omission is consequential because derivative substitution is a primary empirical margin. The model’s narrower role is to identify the campaigns most likely to substitute: under the doubling calibration, $\{\text{RTD}_i > d\}$.

The resulting specification has one private type, quadratic accumulation cost, an exogenously garbled filing, a free-entry bidder cutoff, and a bargaining share disciplined by one appropriability parameter. Any additional mechanism would require removing an existing one rather than expanding this core.

5 Empirical Design

5.1 Exposure and estimating design

The reform has a single adoption date and no untreated group: every initial Schedule 13D filer became subject to the five-business-day deadline on 5 February 2024 (U.S. Securities and Exchange Commission 2023). Identification therefore uses cross-sectional variation in how tightly the common deadline constrains each target, rather than a treated-versus-control comparison.

The required-trading-days measure in equation (1) is constructed from shares outstanding and average daily volume over the six months ending 60 trading days before the trigger. It is defined for every filing, including one-time filers, and does not use the campaign’s realised stake or outcome. The lag reduces contamination from the activist’s own trading. I will diagnose any remaining contamination using the Item 5(c) transaction tables and repeat the analysis with a 120-trading-day lag, following the concern in Collin-Dufresne and Fos (2015) that a filer’s own trading can contaminate liquidity measures.

Under the model, the post-threshold increment is capped at $0.05d/\text{RTD}_i$. At the doubling calibration, in which the desired increment is five percentage points, the cap binds when $\text{RTD}_i > d$. The continuous measure is the primary treatment intensity; the binary indicator is used only to interpret that calibration.

The identifying assumption is that, absent the reform, outcomes at different levels of required trading days would have followed parallel paths and that the dose-response relationship would

have remained stable over the estimation window. A linear specification provides a summary:

$$Y_{i,q} = \alpha_{a(i),q} + \gamma_{\ell(i),q} + \beta \text{Post}_q \times \text{RTD}_i + \varepsilon_{i,q}, \quad (12)$$

where $\alpha_{a(i),q}$ denotes activist-by-quarter fixed effects and $\gamma_{\ell(i),q}$ denotes liquidity-decile-by-quarter fixed effects. The full specification also includes size-decile-by-quarter fixed effects. Since β is a weighted average when responses are heterogeneous, the primary estimates are non-parametric across exposure bins:

$$Y_{i,q} = \alpha_{a(i),q} + \gamma_{\ell(i),q} + \sum_k \beta_k \mathbf{1}[\text{RTD-bin}_i = k] \times \text{Post}_q + \varepsilon_{i,q}. \quad (13)$$

Activist-by-quarter effects absorb fund flows and changes in an activist’s strategy. Liquidity- and size-decile-by-quarter effects absorb time-varying differences across the target distribution.

5.2 Identification threats

Anticipation. The amendments were proposed on 10 March 2022, the comment period reopened on 28 April 2023, and the rule was adopted on 10 October 2023, four months before its effective date (U.S. Securities and Exchange Commission 2023). The main analysis excludes the period from October 2023 through February 2024 from the pre-reform sample. Robustness tests also exclude windows around the March 2022 proposal and the April 2023 reopening.

Bundled rule changes. Release 33-11253 shortened the initial Schedule 13D deadline, shortened the deadline for Schedule 13D amendments to two business days, restructured Schedule 13G deadlines, and required structured XML filing from 18 December 2024 (U.S. Securities and Exchange Commission 2023). It also declined to adopt proposed Rule 13d-3(e) and clarified Item 6 derivative disclosure without expanding its scope. The amendment rule mechanically changes post-filing accumulation, so that outcome is excluded from the behavioural tests. The primary outcomes are trigger-to-filing accumulation, stake at filing, campaign entry and composition, and Item 6 derivative language. I will also report switching between Schedules 13G and 13D among holders near five per cent. A contemporaneous shift in that rate would indicate that the Schedule 13G changes, rather than the initial Schedule 13D deadline, affect outcomes defined over the pooled population.

Exposure and target characteristics. Required trading days depend on target size and liquidity by construction. The SEC’s Table 6 shows why realised accumulation cannot substitute for predetermined exposure. Campaigns that had accumulated less than 90 per cent of their reported stake by the amended deadline involved larger and more prominent activists and larger, more liquid targets. Their average target market capitalisation was \$1.8 billion rather than \$916 million; their Amihud ratio was 0.09 rather than 0.13; the prominent-activist share was 43.6 rather than 29.8 per cent; their filing-window cumulative abnormal return was 17.2 rather than 5.7 per cent; and their average increase in shareholder value was \$222 million rather than \$36 million (U.S. Securities and Exchange Commission 2023). I will report covariate balance across exposure bins before presenting estimates and include the fixed effects described above (Collin-Dufresne and Fos 2015; Collin-Dufresne and Fos 2016; Edmans et al. 2013).

An activist-level exposure measure based on past campaigns is retained only as a heterogeneity split. It cannot identify the main effect because most filers appear too rarely. Polk et al.

(2024) report 9,685 initial filings by 5,863 entities from 2001–2022, or 1.65 filings per entity. In the project’s initial 300-filing sample, 259 filer CIKs are unique, only 29 filers appear at least twice, and only three appear at least three times. Requiring pre- and post-reform campaigns would therefore restrict the analysis to a small group of repeat activists that are unusually able to pre-position or use derivatives.

5.3 Outcomes and statistical power

The SEC’s own tables imply a small population-average treatment. Only 3 per cent of non-corporate-action campaigns, about seven per year, had accumulated less than 90 per cent of their reported stake by the amended deadline; 1 per cent, about one per year, had accumulated less than 75 per cent (U.S. Securities and Exchange Commission 2023). Among the 20 per cent that continued to accumulate after the deadline, only 5.9 per cent of the reported stake was acquired within the filing window on average. Together, these figures imply an average accumulation effect on the order of one tenth of a percentage point of shares outstanding. The interpretation of any estimate will remain bounded by that scale.

The expected post-reform sample contains roughly 540–700 non-corporate-action campaigns from 2024–2026, before any repeat-activist restriction. Table 1 separates outcomes that this sample can reasonably inform from outcomes that are descriptive.

Table 1: Outcome hierarchy and statistical power

Outcome	Role	Interpretation and power
Entry and target composition	primary	Best-powered extensive margin; not selected by conditioning on a filing outcome.
Stake at filing and trigger-to-filing accumulation	primary	Campaign-level tests of the deadline’s accumulation effect.
Item 6 derivative language	primary	Campaign-level test of substitution through the rule’s treatment of cash-settled derivatives.
Filing delay	descriptive	Mechanical compliance outcome; not evidence of a behavioural response.
Filing return and pre-filing drift	secondary	Moderately powered and dependent on CRSP access.
Six- and twelve-month bid hazard	descriptive	Confidence intervals only. The sale-intent triple interaction has a minimum detectable effect of 17–31 percentage points on a base rate of about 17 percentage points, with roughly three bids in the materially exposed post-reform cell.

5.4 EDGAR first-stage analysis

The first-stage analysis uses only EDGAR data and will be pre-registered against minimum detectable effects calculated from the SEC’s Tables 3, 5, and 6. It contains five components:

1. untrimmed filing-delay distributions, reported separately for corporate-action and non-corporate-action filings;
2. pre- and post-reform distributions of the stake reported at filing;

3. entry and target composition by a coarse, price-free proxy for required trading days until CRSP-based average daily volume is available;
4. the prevalence of Item 6 cash-settled-derivative language; and
5. switching between Schedules 13G and 13D near the five-per-cent threshold.

Filing-delay compression and accumulation after the legal deadline are mechanical consequences of compliance and cannot establish a behavioural response.

5.5 Inference and falsification

The principal threat is a continuous-dose estimate driven by a particular placebo date, exposure assignment, or macroeconomic cycle. I will compare the estimated reform effect with distributions generated from placebo reform dates between 2015 and 2023 and from permutations of required trading days across activists. Quarterly event studies by exposure bin begin in 2019, and a joint pre-trend test is required before a causal interpretation. A non-US activist block sample based on UK DTR5 and E.U. Transparency Directive notifications provides a control for the global merger and interest-rate cycle. It will not be used as a legal-threshold comparison because the disclosure regimes are not otherwise comparable.

5.6 The timing of target gains

Total target gain is divided into the filing return $G_{\text{fil}} = \log(P^F/P^0)$, the post-filing run-up $G_{\text{run}} = \log(P^B/P^F)$, and the bid markup $G_{\text{mkp}} = \log(B/P^B)$. The unaffected price P^0 is measured 60 trading days before the trigger in both reform periods, with robustness at 30 and 120 days. Fixing this anchor prevents the treatment from moving the starting point of the decomposition.

The reform shortened the median trigger-to-filing interval from seven to five business days in the project’s initial data. This mechanically shortens the calendar span in the filing return and lengthens the span in the run-up, even if behaviour does not change. The primary filing measure is therefore the filing-date $[-1, +1]$ cumulative abnormal return as a share of total gain, with the trigger-to-bid horizon held fixed. A mechanical placebo recuts pre-reform campaigns as though they had filed under a five-business-day rule. A behavioural placebo restricts the sample to campaigns that already filed within five business days under the old rule, for which the reform did not change the legal window in practice.

If an M&A outcomes database is unavailable, I will hand-collect the acquired-target sub-sample from Forms 8-K and DEFM14A using standard conventions for toeholds, premia, and deal outcomes (Betton et al. 2014; Eckbo et al. 2025). Cross-sectional tests use industry M&A intensity as a proxy for fringe-bidder arrival, the strategic-buyer share as a proxy for the portability of activist improvements, and a discontinuity check around the five-per-cent ownership threshold (Corum and Levit 2019; Burkart and Lee 2022; Brav et al. 2008). The capitalisation interpretation remains a measurement restriction, not a separate contribution (Celentano and Levine 2025).

5.7 Data feasibility

The core sample spans 2022–2026. This window brackets the reform while limiting the Item 5(c) extraction burden; the SEC’s published 2011–2021 tables provide a pre-period benchmark.

EDGAR supports the first-stage analysis. CRSP or WRDS access is required for average daily volume and return-based outcomes, and an M&A source is required for the full gain decomposition. These access decisions govern the price-based extensions, not the EDGAR analysis. The companion execution plan records the acquisition schedule, validation thresholds, and fallbacks.

6 Literature and Position

6.1 Disclosure rules and stake accumulation

Mandatory disclosure affects both the cost of acquiring a position and the incentive to monitor or seek control. Grossman and Hart (1980a) identify the basic trade-off: disclosure protects target shareholders but makes stealth acquisition more costly. Ordonez-Calafi and Bernhardt (2022) analyse an ownership threshold in a one-shot Kyle-style setting. Their policy instrument is a static stake-size trigger, not the period between threshold crossing and mandatory filing. Their discussion of objective-specific thresholds, including sale-seeking campaigns, leaves open the corresponding question for filing deadlines.

Bonetti et al. (2020) provide the closest empirical disclosure design. They exploit the staggered implementation of the E.U. Transparency Directive from 2007 to 2009 and find that tighter disclosure reduces toeholds and control acquisitions, while moving target and acquirer announcement returns in opposite directions. Their setting concerns ordinary corporate bidders, not activist Schedule 13D campaigns, and combines a lower ownership threshold with a shorter notification period. It therefore does not isolate the filing window. Their bidding-competition mechanism emphasises deterrence: disclosure can invite competing bids and increase an acquirer’s cost (Fishman 1988). In the present setting, disclosure may instead facilitate bidder search by identifying a target as available for sale. Whether certification or deterrence dominates is an empirical question, with implications for heterogeneity by sale intent, asset opacity, the set of natural buyers, and industry merger activity.

The strategic-trading literature supplies the basis for the reduced-form accumulation cost. Back et al. (2018) endogenise activist trading and liquidity, while Collin-Dufresne and Fos (2016) model trading on long-lived information against stochastic liquidity. A quadratic cost captures their level prediction that a longer trading horizon and greater liquidity reduce the marginal cost of a given position. It abstracts from endogenous intra-window timing and path-dependent liquidity. Collin-Dufresne and Fos (2015) show that Schedule 13D filers trade on days when measured price impact is almost 30 per cent below its sample average, cautioning against treating conventional pre-filing liquidity measures as uncontaminated. The broader literature on exit, voice, and liquidity provides the background (Maug 1998; Kahn and Winton 1998; Edmans et al. 2013; Edmans and Holderness 2017), but does not study a disclosure window or the timing of takeover gains.

6.2 Activism, control, and structural evidence

Burkart and Lee (2022) model activism as intermediation between an underperforming target and the market for corporate control. Their activist brokers a sale after a proxy-style campaign, using board control to mitigate the Grossman–Hart and Jensen–Meckling free-rider problems. The toehold is exogenous, however, and the model is entirely post-disclosure. The acquisition of the toehold in anonymous predisclosure markets is left to future work.

Corum and Levit (2019) model secondary-market inference and disclosure of activist ownership. An activist trades against a Kyle (1985) market maker, and ownership becomes public through a Schedule 13D publication step. That step is instantaneous and is not governed by a disclosure threshold or filing deadline. Their model nevertheless supplies the bidder-discovery mechanism used here: an activist can signal that a target is available for sale. It also shows that a credible proxy threat can raise takeover probability while reducing the conditional premium through weaker board bargaining. The present paper studies the filing window, which their model does not vary.

Cetemen et al. (2026) analyse leader-follower trading and wolf-pack accumulation in continuous time. They mention the February 2024 acceleration as institutional background, but do not make the window a choice or policy variable. The structural literature likewise quantifies campaigns without varying disclosure timing. Johnson and Swem (2021) estimate a dynamic reputation model in which accumulation is subsumed in a lognormal campaign-cost draw and filing coincides with campaign initiation. Albuquerque et al. (2022) decompose the average Schedule 13D announcement return of 6.34 per cent into treatment, stock-picking, and sample-selection components of 75.2, 12.2, and 12.6 per cent. Their activism cost is a fixed scalar, and they do not study takeover premia.

Celentano and Levine (2025) estimate a simulated-method-of-moments model on 2001–2019 data. They report that activist presence reduces the conditional bid premium by 13.7 per cent, or 5.2 percentage points, because it weakens the board’s bargaining position. They also argue that the pre-announcement price incorporates the activist’s signal about a potential acquirer. A lower measured premium therefore need not imply a lower total gain to target shareholders. Their model treats the activist’s decision and information as instantaneously public and contains no disclosure timing. Finally, Gantchev (2013) estimate an average campaign cost of about \$10.5 million, roughly half in the proxy stage. Those legal, advisory, and proxy-fight expenditures differ from the market-impact cost of building a predisclosure stake used here.

6.3 Takeover gains

The standard takeover decomposition separates a pre-bid run-up from an announcement markup. Schwert (1996) measures the run-up from day -42 to day -1 and finds little substitution between the run-up and markup, with at least two-thirds of the run-up added to the total price. Betton et al. (2014) find materially stronger substitution after adjusting the run-up for changes in the target’s stand-alone value: the estimated slope changes from -0.09 under the unadjusted method to -0.39 . Eckbo et al. (2025) summarise the conventional event windows, including a run-up from day -41 to day -2 and a separate three-day announcement window. The decomposition in this paper adds the activist filing as a distinct event and tests the joint movement of the filing return, run-up, and markup.

Activism is closely associated with takeovers. Greenwood and Schor (2009) show that acquisition outcomes account for much of the abnormal return to Schedule 13D activism and estimate that activism raises takeover probability by about eleven percentage points. Brav et al. (2008) report that selling the company is an objective in roughly 14–18 per cent of campaigns and that about 15 per cent of activist targets are acquired. Boyson et al. (2017) find that activist targets are more likely to receive bids and merge. That last characterisation rests on an author-posted summary rather than the primary text, so it is used only for this narrow point and does not support the sale-intent codebook. None of these studies examines disclosure timing.

Table 2: Position in the literature

	Filing window	Realised reform	Gain anatomy	Schedule 13D
Ordonez-Calafi and Bernhardt (2022)	no	no	no	yes
Burkart and Lee (2022)	no	no	no	yes
Corum and Levit (2019)	no	no	no	yes
Johnson and Swem (2021)	no	no	no	yes
Albuquerque et al. (2022)	no	no	partial	yes
Celentano and Levine (2025)	no	no	partial	yes
Bonetti et al. (2020)	partial	no	no	no
Polk et al. (2024)	partial	no	no	yes
Cetemen et al. (2026)	no	no	no	yes
<i>This paper</i>	yes	yes	yes	yes

6.4 The 2024 reform and the paper’s position

The SEC’s modernisation of beneficial-ownership reporting shortened the initial Schedule 13D deadline, shortened amendment deadlines, and introduced structured XML filing (U.S. Securities and Exchange Commission 2023). It declined to adopt proposed Rule 13d-3(e), instead clarifying Item 6 derivative disclosure. February 2025 guidance further addresses when calls for a sale or restructuring, alternative director nominees, or conditional voting support make the shorter Schedule 13G unavailable (U.S. Securities and Exchange Commission, Division of Corporation Finance 2025).

The SEC’s baseline indicates that the new deadline bound a minority of campaigns. Roughly 29 per cent of 2022 initial filings, or 41 per cent of timely filings, were made within five business days; about 80 per cent of filers had completed accumulation by then; and 97 per cent had reached 90 per cent of their reported stake. The SEC nevertheless projected approximately \$810 million in annual foregone shareholder value.

Polk et al. (2024) is the only prior study focused on the acceleration. It uses 9,685 initial Schedule 13D filings from 2001–2022 and compares investors who filed voluntarily within five days with those who waited the full ten. Its outcomes are returns and trading volume around trigger and filing dates. The sample ends before the reform and contains no campaign-resolution, bidder-entry, takeover, or premium outcomes. A systematic search conducted on 2 August 2026 across SSRN, NBER, arXiv, SEC and DERA materials, forthcoming journal pages, and author websites found no study of realised post-February-2024 Schedule 13D outcomes. The closest related work, Celentano and Levine (2025), neither studies the reform nor uses post-reform data.

The paper’s contribution is therefore specific. It treats the filing window as a time constraint on post-threshold stake formation and evaluates realised responses to the 2024 rule. The activist-as-intermediary mechanism and the capitalisation interpretation of lower conditional premia are established in the cited literature; they motivate, rather than constitute, the contribution. Table 2 summarises the distinction.

7 Interpretation of Possible Findings

The two primary behavioural margins are the response of campaign entry and composition to required-trading-days exposure and the response of Item 6 derivative use. Table 3 states the interpretation of each combination without presuming an outcome.

Table 3: Interpretation of the primary empirical margins

	No increase in Item 6 derivatives	Increase in Item 6 derivatives
Entry or composition responds to exposure	Campaign supply or target composition changes without detectable derivative substitution.	Campaign activity changes and derivative use rises in exposed targets.
No detectable entry or composition response	No detectable behavioural response within the powered range.	Derivative use rises without a detectable change in campaign supply or target composition.

Subject to the identification checks, a campaign response without derivative substitution validates the chilling channel in direction, though not necessarily in magnitude. A joint response links the accumulation constraint to the regulatory treatment of derivatives. An increase in derivative use without a campaign response would identify the decision not to adopt proposed Rule 13d-3(e), rather than the headline deadline change, as the operative policy margin. No response on either margin is the modal ex ante outcome given the SEC’s compliance statistics and would provide a policy null bounded by the reported confidence intervals and minimum detectable effects.

These interpretations remain conditional on the pre-trend, placebo, and bundled-reform checks. A null estimate does not establish that the reform had no effect outside the detectable range. Conversely, a response in a mechanically affected timing measure does not establish a behavioural effect.

The gain decomposition addresses a separate measurement question. It identifies whether returns associated with activist targets are realised at filing, during the pre-bid run-up, or in the bid markup. A lower markup is not by itself evidence of a lower total gain. The relevant test considers all three components over a fixed trigger-to-bid horizon and distinguishes a redistribution of gains across event time from a change in total target-shareholder value.

A Model Derivations

A.1 Accumulation cost

At daily rate r , the dealer’s excess cost is

$$\int_0^r \lambda_L u \, du = \frac{\lambda_L}{2} r^2.$$

The activist minimises $\sum_{t=1}^d (\lambda_L/2) r_t^2$ subject to $\sum_{t=1}^d r_t = x$. Convexity and the symmetric constraint imply $r_t = x/d$. Hence

$$d \frac{\lambda_L}{2} \left(\frac{x}{d} \right)^2 = \frac{\lambda_L}{2} \frac{x^2}{d}.$$

Differentiation gives equation (3) and the degree-minus-one homogeneity in d .

A.2 Stake comparative static

In the unconstrained region, the first-order condition is $b_x - C_x = 0$. Applying the implicit function theorem gives

$$\frac{d\Delta x^*}{dd} = -\frac{b_{xd} - C_{xd}}{b_{xx} - C_{xx}}.$$

Strict concavity makes the denominator negative, so the derivative is positive if and only if $b_{xd} > C_{xd}$. At the constrained corner, $\Delta x_i^* = \phi \text{ADV}_i d / S_{\text{out},i}$, and direct differentiation gives

$$\frac{d\Delta x_i^*}{dd} = \frac{\phi \text{ADV}_i}{S_{\text{out},i}} > 0.$$

References

- Albuquerque, Rui, Vyacheslav Fos, and Enrique Schroth (2022). “Value Creation in Shareholder Activism”. In: *Journal of Financial Economics* 145.2, pp. 153–178.
- Back, Kerry et al. (2018). “Activism, Strategic Trading, and Liquidity”. In: *Econometrica* 86.4, pp. 1431–1463.
- Betton, Sandra et al. (2014). “Merger Negotiations with Stock Market Feedback”. In: *Journal of Finance* 69.4, pp. 1705–1745.
- Bonetti, Pietro, Miguel Duro, and Gaizka Ormazabal (2020). “Disclosure Regulation and Corporate Acquisitions”. In: *Journal of Accounting Research* 58.1, pp. 55–103.
- Boyson, Nicole M., Nikolay Gantchev, and Anil Shivdasani (2017). “Activism Mergers”. In: *Journal of Financial Economics* 126.1, pp. 54–73.
- Brav, Alon et al. (2008). “Hedge Fund Activism, Corporate Governance, and Firm Performance”. In: *Journal of Finance* 63.4, pp. 1729–1775.
- Burkart, Mike and Samuel Lee (2022). “Activism and Takeovers”. In: *Review of Financial Studies* 35.4, pp. 1868–1896.
- Celentano, Francesco and Oliver Levine (2025). “Shareholder Activism, Takeovers, and Managerial Discipline”. R&R, Review of Financial Studies. Swiss Finance Institute Research Paper No. 25-81; SSRN 5506659.
- Cetemen, Doruk et al. (2026). “Leader-Follower Dynamics in Shareholder Activism”. In: *Journal of Finance*. Forthcoming.
- Collin-Dufresne, Pierre and Vyacheslav Fos (2015). “Do Prices Reveal the Presence of Informed Trading?” In: *Journal of Finance* 70.4, pp. 1555–1582.
- (2016). “Insider Trading, Stochastic Liquidity, and Equilibrium Prices”. In: *Econometrica* 84.4, pp. 1441–1475.
- Corum, Adrian Aycan and Doron Levit (2019). “Corporate Control Activism”. In: *Journal of Financial Economics* 133.1, pp. 1–17.
- Eckbo, B. Espen, Nadya Malenko, and Karin S. Thorburn (2025). *Corporate Takeovers*. ECGI Finance Working Paper No. 1030/2025. European Corporate Governance Institute.
- Edmans, Alex, Vivian W. Fang, and Emanuel Zur (2013). “The Effect of Liquidity on Governance”. In: *Review of Financial Studies* 26.6, pp. 1443–1482.
- Edmans, Alex and Clifford G. Holderness (2017). “Blockholders: A Survey of Theory and Evidence”. In: *The Handbook of the Economics of Corporate Governance*. Ed. by Benjamin E. Hermalin and Michael S. Weisbach. Vol. 1. North-Holland, pp. 541–636.
- Fishman, Michael J. (1988). “A Theory of Preemptive Takeover Bidding”. In: *RAND Journal of Economics* 19.1.

- Gantchev, Nickolay (2013). “The Costs of Shareholder Activism: Evidence from a Sequential Decision Model”. In: *Journal of Financial Economics* 107.3, pp. 610–631.
- Greenwood, Robin and Michael Schor (2009). “Investor Activism and Takeovers”. In: *Journal of Financial Economics* 92.3, pp. 362–375.
- Grossman, Sanford J. and Oliver D. Hart (1980a). “Disclosure Laws and Takeover Bids”. In: *Journal of Finance* 35.2, pp. 323–334.
- (1980b). “Takeover Bids, the Free-Rider Problem, and the Theory of the Corporation”. In: *Bell Journal of Economics* 11.1, pp. 42–64.
- Johnson, Travis L. and Nathan Swem (2021). “Reputation and Investor Activism: A Structural Approach”. In: *Journal of Financial Economics* 139.1, pp. 29–56.
- Kahn, Charles and Andrew Winton (1998). “Ownership Structure, Speculation, and Shareholder Intervention”. In: *Journal of Finance* 53.1, pp. 99–129.
- Kyle, Albert S. (1985). “Continuous Auctions and Insider Trading”. In: *Econometrica* 53.6, pp. 1315–1335.
- Maug, Ernst (1998). “Large Shareholders as Monitors: Is There a Trade-Off between Liquidity and Control?” In: *Journal of Finance* 53.1, pp. 65–98.
- Ordóñez-Calafi, Guillem and Dan Bernhardt (2022). “Blockholder Disclosure Thresholds and Hedge Fund Activism”. In: *Journal of Financial and Quantitative Analysis* 57.7, pp. 2834–2859.
- Polk, Ryan Christopher et al. (2024). “Shrinking the 13D Disclosure Window Will Benefit Non-Activist Investors”. In: *Journal of Financial Regulation and Compliance* 32.4, pp. 516–538.
- Schwert, G. William (1996). “Markup Pricing in Mergers and Acquisitions”. In: *Journal of Financial Economics* 41.2, pp. 153–192.
- U.S. Securities and Exchange Commission (Oct. 2023). *Modernization of Beneficial Ownership Reporting*. Release Nos. 33-11253; 34-98704. Adopted October 2023; effective February 5, 2024. U.S. Securities and Exchange Commission. URL: <https://www.sec.gov/files/rules/final/2023/33-11253.pdf>.
- U.S. Securities and Exchange Commission, Division of Corporation Finance (2025). *Exchange Act Sections 13(d) and 13(g) and Regulation 13D-G Beneficial Ownership Reporting Compliance and Disclosure Interpretations*. Question 103.12 (Feb. 11, 2025; page last updated July 9, 2026). URL: <https://www.sec.gov/divisions/corpfin/guidance/13d-g-interps.htm> (visited on 08/02/2026).